Machine Learning

Lecture 4B: Solved Classroom Copy Through Projection

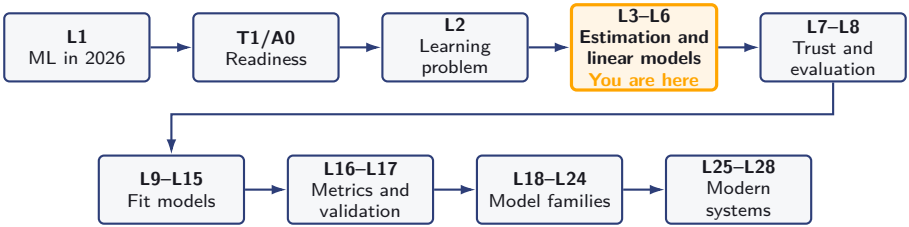
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Civil Engineering; Computer Science and Engineering

Semester I, 2026



Course Roadmap: You Are Here



This solved classroom copy records what we completed through the **algebraic** and **geometric** routes: basis functions, the design matrix, least squares, normal equations, perpendicular residuals, and the projection matrix. The probability and optimisation routes begin in the next class.

Where Lecture 4A Stopped

For each observation,

$$\widehat{y}_i = f_\theta(\boldsymbol{x}_i), \qquad r_i(\boldsymbol{\theta}) = y_i - \widehat{y}_i.$$

The fitted parameters are produced by an estimator:

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} \ell(r_i(\boldsymbol{\theta}))\,.$$

Question

The question left in the room was: if the errors are not normal, what objective should we use?

Today we answer it, then examine the same fit through algebra, geometry, probability, and optimization.

Learning Objectives

By the end of the lecture, you should be able to:

- map an error density to a loss using negative log-likelihood;
- construct a design matrix from basis functions;
- write least squares as a nearest-point problem in $\mathrm{Col}(X)$;
- derive the normal equations and explain why the residual is perpendicular to the model space;
- interpret $P_X \boldsymbol{y}$ and $(I - P_X)\boldsymbol{y}$ as the representable and unrepresentable parts of the response.

Companion interactive laboratory: `ml2026.floodresq.com/explore/linear-regression`

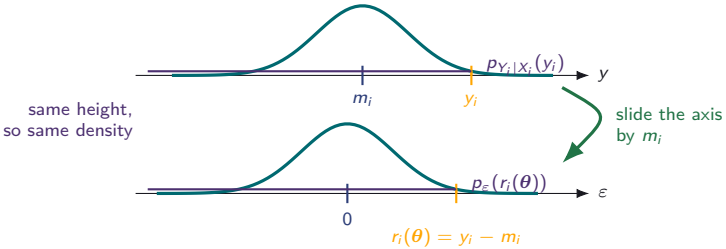
A Translation Connects Response And Error Densities

Fix one input $\boldsymbol{x}_i$ and one candidate parameter vector $\boldsymbol{\theta}$. Define

$$m_i = f_{\boldsymbol{\theta}}(\boldsymbol{x}_i), \qquad y_i = m_i + \varepsilon_i \quad \Longleftrightarrow \quad \varepsilon_i = y_i - m_i = r_i(\boldsymbol{\theta}).$$

This is a translation: subtracting the fixed number m_i changes the origin, not the probability assigned to the corresponding point. Because $|\mathrm{d}(y_i - m_i)/\mathrm{d}y_i| = 1$,

$$\boxed{p_{Y_i|X_i}(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}) = p_{\varepsilon}(y_i - m_i \mid \boldsymbol{x}_i) = p_{\varepsilon}(r_i(\boldsymbol{\theta}) \mid \boldsymbol{x}_i)\,.}$$



From One Residual Density To A Loss

For observation i, the likelihood contribution is now written as

$$p_{Y_i|X_i}(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}) = p_\varepsilon(r_i(\boldsymbol{\theta}) \mid \boldsymbol{x}_i)\,.$$

If observations are conditionally independent given their inputs,

$$p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\theta}) = \prod_{i=1}^{n} p_\varepsilon(r_i(\boldsymbol{\theta}) \mid \boldsymbol{x}_i)\,.$$

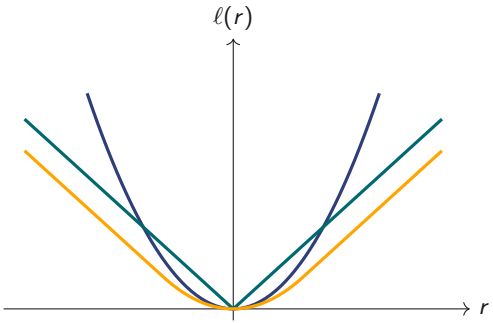
Taking a logarithm turns the product into a sum; multiplying by -1 turns maximization into minimization:

$$\widehat{\boldsymbol{\theta}}_{\mathrm{MLE}} = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} \underbrace{-\log p_\varepsilon(r_i(\boldsymbol{\theta}) \mid \boldsymbol{x}_i)}_{\ell_i(\boldsymbol{\theta})\text{: loss of observation } i}\,.$$

Summary

The error model chooses the likelihood-shaped loss. We next hold the loss idea fixed and specify the predictor $f_{\boldsymbol{\theta}}$ that will generate each residual.

Three Residual Penalties, One Model



Gaussian	$\ell(r) \propto r^2$
Laplace	$\ell(r) \propto \lvert r \rvert$
Huber	quadratic near zero, linear in the tails

Warning

Ordinary least squares can still be computed without normal errors. Normality is what makes squared loss the Gaussian MLE.

Bridge: the loss tells us how to score a mistake; the model tells us which predictions, and therefore which residuals, are possible.

Interactive laboratory: compare error generators, fitted losses, and their minima ↗

What “Linear Model” Means Here

Choose a feature map $\phi(x) = [\phi_0(x), \ldots, \phi_{d-1}(x)]^\top$. The predictor is

$$f_{\boldsymbol{\theta}}(x) = \boldsymbol{\phi}(x)^\top \boldsymbol{\theta} = \sum_{j=0}^{d-1} \theta_j \phi_j(x).$$

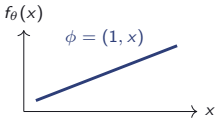
Linear in the parameters

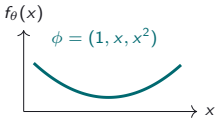
Each θ_j appears to the first power and multiplies a fixed function ϕ_j. There are no terms such as θ_1^2, $\theta_1\theta_2$, or e^{θ_1}.

Straight in x $\quad \theta_0 + \theta_1 x$

Curved in x $\quad \theta_0 + \theta_1 x + \theta_2 x^2$

The ϕ_j may include $x^2, x^3, \sin x$, or fixed radial functions. These can make $f_\theta(x)$ curved in x without making it nonlinear in θ. In linear regression, “linear” describes how the unknown coefficients enter.





Stack The Rows: The Design Matrix Appears

Use one dataset throughout the geometric argument:

$$x = (0, 1, 2), \qquad \boldsymbol{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \qquad \phi(x) = \begin{bmatrix} 1 \\ x \end{bmatrix}.$$

For observation i,

$$\widehat{y}_i = \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}.$$

Stack the three rows. Then multiply the resulting matrix by $\boldsymbol{\theta}$ and check every dimension.

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}}_{X \in \mathbb{R}^{3\times 2}} \underbrace{\begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}}_{\boldsymbol{\theta} \in \mathbb{R}^2} = \underbrace{\begin{bmatrix} \theta_0 \\ \theta_0 + \theta_1 \\ \theta_0 + 2\theta_1 \end{bmatrix}}_{\widehat{\boldsymbol{y}} = X\boldsymbol{\theta} \in \mathbb{R}^3}.$$

$$\boldsymbol{r} = \boldsymbol{y} - X\boldsymbol{\theta} \in \mathbb{R}^3.$$

Each *row* builds one prediction. Each *column* records one feature pattern across all observations.

What Does “Span” Mean?

For the running matrix, name its columns:

$$X = [\boldsymbol{c}_0\ \boldsymbol{c}_1], \qquad \boldsymbol{c}_0 = \begin{bmatrix}1\\1\\1\end{bmatrix}, \quad \boldsymbol{c}_1 = \begin{bmatrix}0\\1\\2\end{bmatrix}.$$

Every parameter choice produces

$$X\boldsymbol{\theta} = \theta_0\boldsymbol{c}_0 + \theta_1\boldsymbol{c}_1.$$

Compute the predictions for $\boldsymbol{\theta} = (1, 0)$, $(0, 1)$, and $(1, \frac{1}{2})$. What remains fixed, and what is allowed to vary?

$$X\begin{bmatrix}1\\0\end{bmatrix} = \boldsymbol{c}_0 = \begin{bmatrix}1\\1\\1\end{bmatrix}, \qquad X\begin{bmatrix}0\\1\end{bmatrix} = \boldsymbol{c}_1 = \begin{bmatrix}0\\1\\2\end{bmatrix},$$

$$X\begin{bmatrix}1\\1/2\end{bmatrix} = \boldsymbol{c}_0 + \tfrac{1}{2}\boldsymbol{c}_1 = \begin{bmatrix}1\\3/2\\2\end{bmatrix}.$$

The columns stay fixed; the coefficients vary. The **span** is simply the set of *all* vectors obtainable as $a\boldsymbol{c}_0 + b\boldsymbol{c}_1$ for real numbers a, b.

$$\mathrm{Col}(X) = \mathrm{span}\{\boldsymbol{c}_0, \boldsymbol{c}_1\}.$$

From Span To A Fitting Problem

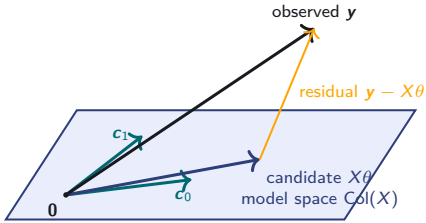
The span of the columns is called the **column space**, abbreviated $\mathrm{Col}(X)$. The model can produce only vectors in

$$\mathrm{Col}(X) = \{X\boldsymbol{\theta} : \boldsymbol{\theta} \in \mathbb{R}^d\} \subseteq \mathbb{R}^n.$$

The observed vector $\boldsymbol{y}$ need not belong to this set. Least squares chooses the allowed prediction with the smallest Euclidean distance:

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \underbrace{\frac{1}{2}\|\boldsymbol{y} - X\boldsymbol{\theta}\|_2^2}_{J(\boldsymbol{\theta})}.$$

Here X and $\boldsymbol{y}$ are fixed; $\boldsymbol{\theta}$ moves.



Next question: what condition identifies the closest point? We derive it rather than assume the right angle.

Interactive laboratory: compare an arbitrary model-space point with the nearest point ↗

What Exactly Is Least Squares Minimizing?

For observation i, the prediction error is vertical in the response coordinate:

$$r_i = y_i - \widehat{y}_i = y_i - \phi(x_i)^\top \boldsymbol{\theta}.$$

Stacking all n errors gives the residual vector $\boldsymbol{r}(\boldsymbol{\theta}) = \boldsymbol{y} - X\boldsymbol{\theta}$. Hence

$$\boxed{J(\boldsymbol{\theta}) = \frac{1}{2}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 = \frac{1}{2}\|\boldsymbol{y} - X\boldsymbol{\theta}\|_2^2.}$$

ℓ_2 notation, unpacked

$$\|\boldsymbol{v}\|_2^2 = \left(\sqrt{v_1^2 + \cdots + v_n^2}\right)^2 = v_1^2 + \cdots + v_n^2.$$

The lower 2 selects Euclidean length; the upper 2 squares it. Here $\boldsymbol{v} = \boldsymbol{r}(\boldsymbol{\theta})$.

In words: least squares adds the squared prediction errors over all n observations; the norm is compact notation for that same sum.

From Least Squares To A Nearest-Point Problem

Because every possible fitted vector has the form $\boldsymbol{z} = X\boldsymbol{\theta} \in \mathrm{Col}(X)$,

$$\widehat{\boldsymbol{y}} = \arg\min_{\boldsymbol{z} \in \mathrm{Col}(X)} \|\boldsymbol{y} - \boldsymbol{z}\|_2^2.$$

Summary

Among all prediction vectors the model can produce, choose the one closest to the observed response vector $\boldsymbol{y}$.

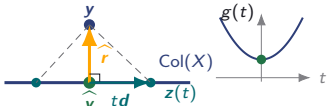
Warning

We square the vertical response errors, not each point's shortest distance to the plotted regression line. The equivalent distance lives in n-dimensional observation space.

Interactive laboratory: move a competing fitted point and compare both squared distances ↗

Why Least Squares Leaves A Perpendicular Residual

Goal: $\widehat{\boldsymbol{r}} = \boldsymbol{y} - \widehat{\boldsymbol{y}} \perp \mathrm{Col}(X)$. **Why move at all?** "Nearest" is a comparison, not yet an equation. So probe it — step off $\widehat{\boldsymbol{y}}$ without leaving the model space, and watch the distance.



1. A legal step. Any $\boldsymbol{d} \in \mathrm{Col}(X)$ gives $\boldsymbol{z}(t) = \widehat{\boldsymbol{y}} + t\boldsymbol{d}$, still a vector the model can produce.
2. Distance along that path.
$g(t) = \|\boldsymbol{y} - \boldsymbol{z}(t)\|_2^2 = \|\widehat{\boldsymbol{r}} - t\boldsymbol{d}\|_2^2$.
3. Nearest puts the bottom at $t = 0$. No competitor beats $\widehat{\boldsymbol{y}}$, so $g(t) \geq g(0)$ for *every* t — that inequality is what "nearest" means. A differentiable function has zero slope at an interior minimum, so $g'(0) = 0$: one equation, in place of infinitely many comparisons.
Board task: expand $g(t)$ and read off what $g'(0) = 0$ forces about $\widehat{\boldsymbol{r}}^\top \boldsymbol{d}$.

$$
\begin{aligned}
g(t) &= \|\boldsymbol{y} - (\widehat{\boldsymbol{y}} + t\boldsymbol{d})\|_2^2 \\
&= \|\widehat{\boldsymbol{r}} - t\boldsymbol{d}\|_2^2 \\
&= (\widehat{\boldsymbol{r}} - t\boldsymbol{d})^\top (\widehat{\boldsymbol{r}} - t\boldsymbol{d}) \\
&= \|\widehat{\boldsymbol{r}}\|_2^2 - 2t\boldsymbol{d}^\top \widehat{\boldsymbol{r}} + t^2 \|\boldsymbol{d}\|_2^2.
\end{aligned}
$$

Therefore

$$g'(t) = -2\boldsymbol{d}^\top \widehat{\boldsymbol{r}} + 2t\|\boldsymbol{d}\|_2^2.$$

Since $t = 0$ minimizes g,

$$0 = g'(0) = -2\boldsymbol{d}^\top \widehat{\boldsymbol{r}} \quad \Longrightarrow \quad \boldsymbol{d}^\top \widehat{\boldsymbol{r}} = 0.$$

Because $\boldsymbol{d}$ was arbitrary,

$$\boxed{\widehat{\boldsymbol{r}} \perp \mathrm{Col}(X).}$$

Interactive laboratory: reveal the arbitrary fit, nearest fit, and orthogonality in sequence ↗

Route 1 – Algebra Solves The Normal Equations

Start from

$$J(\boldsymbol{\theta}) = \frac{1}{2}(\boldsymbol{y} - X\boldsymbol{\theta})^\top (\boldsymbol{y} - X\boldsymbol{\theta}).$$

Complete four steps:

1. expand the quadratic;
2. differentiate with respect to $\boldsymbol{\theta}$;
3. set the gradient to zero;
4. name the resulting equation.

Check the dimension of every term.

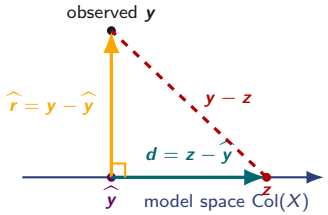
$$J(\boldsymbol{\theta}) = \tfrac{1}{2}\boldsymbol{y}^\top \boldsymbol{y} - \boldsymbol{\theta}^\top X^\top \boldsymbol{y} + \tfrac{1}{2}\boldsymbol{\theta}^\top X^\top X\boldsymbol{\theta},$$
$$\nabla J(\boldsymbol{\theta}) = X^\top (X\boldsymbol{\theta} - \boldsymbol{y}).$$

At the minimizer,

$$\nabla J(\widehat{\boldsymbol{\theta}}) = \mathbf{0} \quad \Longrightarrow \quad \boxed{X^\top X\widehat{\boldsymbol{\theta}} = X^\top \boldsymbol{y}}.$$

These are the **normal equations**. The gradient is in $\mathbb{R}^d$; both sides of the normal equations are in $\mathbb{R}^d$.

Route 2 – Geometry Gives The Nearest Point



Let $z \neq \widehat{y}$ be another allowed fitted vector. Write $d = z - \widehat{y} \in \mathrm{Col}(X)$.
Compare $\|y - z\|_2^2$ with $\|y - \widehat{y}\|_2^2$ without relying only on the picture.

The normal equations recover the same orthogonality condition:

$$X^\top \widehat{r} = 0 \quad \Longrightarrow \quad \widehat{r} \perp \mathrm{Col}(X).$$

Because $d \in \mathrm{Col}(X)$, $\widehat{r}^\top d = 0$. Also, $y - z = y - (\widehat{y} + d) = \widehat{r} - d$. Therefore

$$\begin{aligned} \|y - z\|_2^2 &= \|\widehat{r} - d\|_2^2 \\ &= \|\widehat{r}\|_2^2 - 2\widehat{r}^\top d + \|d\|_2^2 \\ &= \|\widehat{r}\|_2^2 + \|d\|_2^2 \\ &> \|\widehat{r}\|_2^2 \qquad (d \neq 0). \end{aligned}$$

Thus every nonzero sideways move adds avoidable squared distance.

Interactive laboratory: move the alternative point and watch the excess grow as $\|d\|^2$ ↗

From The Parameter Estimate To The Projection Matrix

Assume the columns of X are linearly independent, so $X^\top X$ is invertible.
Solve the normal equations for $\widehat{\boldsymbol{\theta}}$. Then substitute the result into

$$\widehat{\boldsymbol{y}} = X\widehat{\boldsymbol{\theta}}.$$

Identify the matrix that maps any observed response vector $\boldsymbol{y}$ to its closest model-space vector.

$$\widehat{\boldsymbol{\theta}} = (X^\top X)^{-1} X^\top \boldsymbol{y}.$$

Therefore,

$$\begin{aligned}\widehat{\boldsymbol{y}} &= X\widehat{\boldsymbol{\theta}} \\ &= X(X^\top X)^{-1} X^\top \boldsymbol{y} \\ &= P_X \boldsymbol{y},\end{aligned}$$

where

$$\boxed{P_X = X(X^\top X)^{-1} X^\top}.$$

P_X is the orthogonal projection matrix onto $\mathrm{Col}(X)$.

Why P_X Is A Projector

A projection has two defining behaviors:

1. applying it twice changes nothing;
2. an orthogonal projection is symmetric.

Starting from

$$P_X = X(X^\top X)^{-1}X^\top,$$

show that $P_X^2 = P_X$ and $P_X^\top = P_X$.
Then interpret

$$\widehat{\boldsymbol{y}} = P_X\boldsymbol{y}, \qquad \widehat{\boldsymbol{r}} = (I - P_X)\boldsymbol{y}.$$

Idempotence:

$$P_X^2 = X(X^\top X)^{-1} \underbrace{X^\top X}_{\text{middle product}} (X^\top X)^{-1}X^\top$$

$$= P_X.$$

Symmetry:

$$P_X^\top = X\left[(X^\top X)^{-1}\right]^\top X^\top = P_X.$$

So $P_X\boldsymbol{y}$ is the representable part of $\boldsymbol{y}$; $(I - P_X)\boldsymbol{y}$ is the part this feature space cannot represent.

Class Question 1: Where Does d Come From?

Step 1: identify the two fitted-response vectors. The chosen fit and any competing fit are

$$\widehat{\boldsymbol{y}} = X\widehat{\theta}, \qquad \boldsymbol{z} = X\theta'.$$

Both lie in $\mathrm{Col}(X)$.

Step 2: subtract them. Define the displacement from the chosen fit to the competitor:

$$\boxed{\boldsymbol{d} := \boldsymbol{z} - \widehat{\boldsymbol{y}} = X\theta' - X\widehat{\theta} = X(\theta' - \widehat{\theta}).}$$

Therefore $\boldsymbol{d} \in \mathrm{Col}(X)$ and $\boldsymbol{z} = \widehat{\boldsymbol{y}} + \boldsymbol{d}$.

Step 3: introduce a scalar only to move along that displacement.

$$\boldsymbol{z}(t) = \widehat{\boldsymbol{y}} + t\boldsymbol{d}, \qquad \boldsymbol{z}(0) = \widehat{\boldsymbol{y}}, \quad \boldsymbol{z}(1) = \boldsymbol{z}.$$

$\boldsymbol{d}$ is not an additional model parameter. It is simply **competitor minus chosen fit** in observation space.

Class Question 1: Which Residual Depends On d?

Step 1: fix the base-point residual.

$$\widehat{\boldsymbol{r}} := \boldsymbol{y} - \widehat{\boldsymbol{y}}.$$

It is already determined once $\widehat{\boldsymbol{y}}$ has been chosen.

Step 2: calculate the residual while moving along $\boldsymbol{d}$.

$$\begin{aligned} \boldsymbol{r}(t; \boldsymbol{d}) &:= \boldsymbol{y} - \boldsymbol{z}(t) \\ &= \boldsymbol{y} - (\widehat{\boldsymbol{y}} + t\boldsymbol{d}) \\ &= (\boldsymbol{y} - \widehat{\boldsymbol{y}}) - t\boldsymbol{d} \\ &= \widehat{\boldsymbol{r}} - t\boldsymbol{d}. \end{aligned}$$

Thus the *moving* residual depends on t and $\boldsymbol{d}$. At the base point,

$$\boxed{\boldsymbol{r}(0; \boldsymbol{d}) = \widehat{\boldsymbol{r}},}$$

which is the same vector for every competing direction.

Class Question 1: Differentiate Along That Direction

Step 1: make the multivariable objective a one-variable function.

$$g(t) := \frac{1}{2}\|\boldsymbol{y} - (\widehat{\boldsymbol{y}} + t\boldsymbol{d})\|_2^2 = \frac{1}{2}(\widehat{\boldsymbol{r}} - t\boldsymbol{d})^T(\widehat{\boldsymbol{r}} - t\boldsymbol{d}).$$

Step 2: expand, then differentiate with respect to t.

$$g(t) = \tfrac{1}{2}\widehat{\boldsymbol{r}}^T\widehat{\boldsymbol{r}} - t\widehat{\boldsymbol{r}}^T\boldsymbol{d} + \tfrac{1}{2}t^2\boldsymbol{d}^T\boldsymbol{d},$$
$$g'(t) = -\widehat{\boldsymbol{r}}^T\boldsymbol{d} + t\boldsymbol{d}^T\boldsymbol{d}.$$

Step 3: evaluate at the fitted point $t = 0$.

$$\boxed{g'(0) = -\widehat{\boldsymbol{r}}^T\boldsymbol{d} = 0 \quad \text{for every } \boldsymbol{d} \in \mathrm{Col}(X).}$$

Therefore $\widehat{\boldsymbol{r}} \perp \mathrm{Col}(X)$. We hold $\boldsymbol{d}$ fixed while differentiating; it remains present in the derivative.

Class Question 2: Expand The Transpose Before Differentiating

Step 1: transpose the residual carefully.

$$(\boldsymbol{y} - X\boldsymbol{\theta})^T = \boldsymbol{y}^T - (X\boldsymbol{\theta})^T = \boldsymbol{y}^T - \boldsymbol{\theta}^T X^T.$$

Step 2: multiply the two factors in their original order.

$$\begin{aligned} J(\boldsymbol{\theta}) &= \tfrac{1}{2}(\boldsymbol{y}^T - \boldsymbol{\theta}^T X^T)(\boldsymbol{y} - X\boldsymbol{\theta}) \\ &= \tfrac{1}{2}\left(\boldsymbol{y}^T\boldsymbol{y} - \boldsymbol{y}^T X\boldsymbol{\theta} - \boldsymbol{\theta}^T X^T\boldsymbol{y} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}\right). \end{aligned}$$

Step 3: combine the middle terms. The quantity $\boldsymbol{y}^T X\boldsymbol{\theta}$ is 1×1, hence equals its transpose:

$$\boldsymbol{y}^T X\boldsymbol{\theta} = (\boldsymbol{y}^T X\boldsymbol{\theta})^T = \boldsymbol{\theta}^T X^T\boldsymbol{y}.$$

Therefore

$$\boxed{J(\boldsymbol{\theta}) = \tfrac{1}{2}\boldsymbol{y}^T\boldsymbol{y} - \boldsymbol{\theta}^T X^T\boldsymbol{y} + \tfrac{1}{2}\boldsymbol{\theta}^T(X^T X)\boldsymbol{\theta}.}$$

Class Question 2: Derive The Quadratic Rule

Let $A = X^T X$ and

$$q(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^T A \boldsymbol{\theta} = \frac{1}{2}\sum_{j=1}^{d}\sum_{k=1}^{d}\theta_j A_{jk}\theta_k.$$

Differentiate with respect to one coordinate θ_m. It can appear as the first factor ($j = m$) or the second factor ($k = m$):

$$\begin{aligned}\frac{\partial q}{\partial \theta_m} &= \frac{1}{2}\sum_{k=1}^{d} A_{mk}\theta_k + \frac{1}{2}\sum_{j=1}^{d}\theta_j A_{jm} \\ &= \left[\frac{1}{2}(A + A^T)\boldsymbol{\theta}\right]_m.\end{aligned}$$

Hence

$$\nabla_{\boldsymbol{\theta}} q = \frac{1}{2}(A + A^T)\boldsymbol{\theta}.$$

Since $A = X^T X$ is symmetric, this becomes $A\boldsymbol{\theta} = X^T X\boldsymbol{\theta}$. Therefore

$$\boxed{\nabla_{\boldsymbol{\theta}} J = -X^T \boldsymbol{y} + X^T X\boldsymbol{\theta} = X^T(X\boldsymbol{\theta} - \boldsymbol{y}).}$$

Class Question 2: Check The Answer With Differentials

Step 1: abbreviate the residual.

$$\boldsymbol{r} = \boldsymbol{y} - X\boldsymbol{\theta}, \qquad J = \tfrac{1}{2}\boldsymbol{r}^T\boldsymbol{r}, \qquad \mathrm{d}\boldsymbol{r} = -X\,\mathrm{d}\boldsymbol{\theta}.$$

Step 2: differentiate the scalar product.

$$\begin{aligned}\mathrm{d}J &= \tfrac{1}{2}\left[(\mathrm{d}\boldsymbol{r})^T\boldsymbol{r} + \boldsymbol{r}^T\mathrm{d}\boldsymbol{r}\right] \\ &= \boldsymbol{r}^T\mathrm{d}\boldsymbol{r} = -\boldsymbol{r}^T X\,\mathrm{d}\boldsymbol{\theta} = (-X^T\boldsymbol{r})^T\mathrm{d}\boldsymbol{\theta}.\end{aligned}$$

Step 3: match the coefficient of $\mathrm{d}\boldsymbol{\theta}$. Because $\mathrm{d}J = (\nabla_{\boldsymbol{\theta}} J)^T\mathrm{d}\boldsymbol{\theta}$,

$$\boxed{\nabla_{\boldsymbol{\theta}} J = -X^T\boldsymbol{r} = X^T(X\boldsymbol{\theta} - \boldsymbol{y}).}$$

Dimension check: $X^T \in \mathbb{R}^{d\times n}$ and $\boldsymbol{r} \in \mathbb{R}^n$, so $X^T\boldsymbol{r} \in \mathbb{R}^d$, matching a gradient with respect to $\boldsymbol{\theta} \in \mathbb{R}^d$.